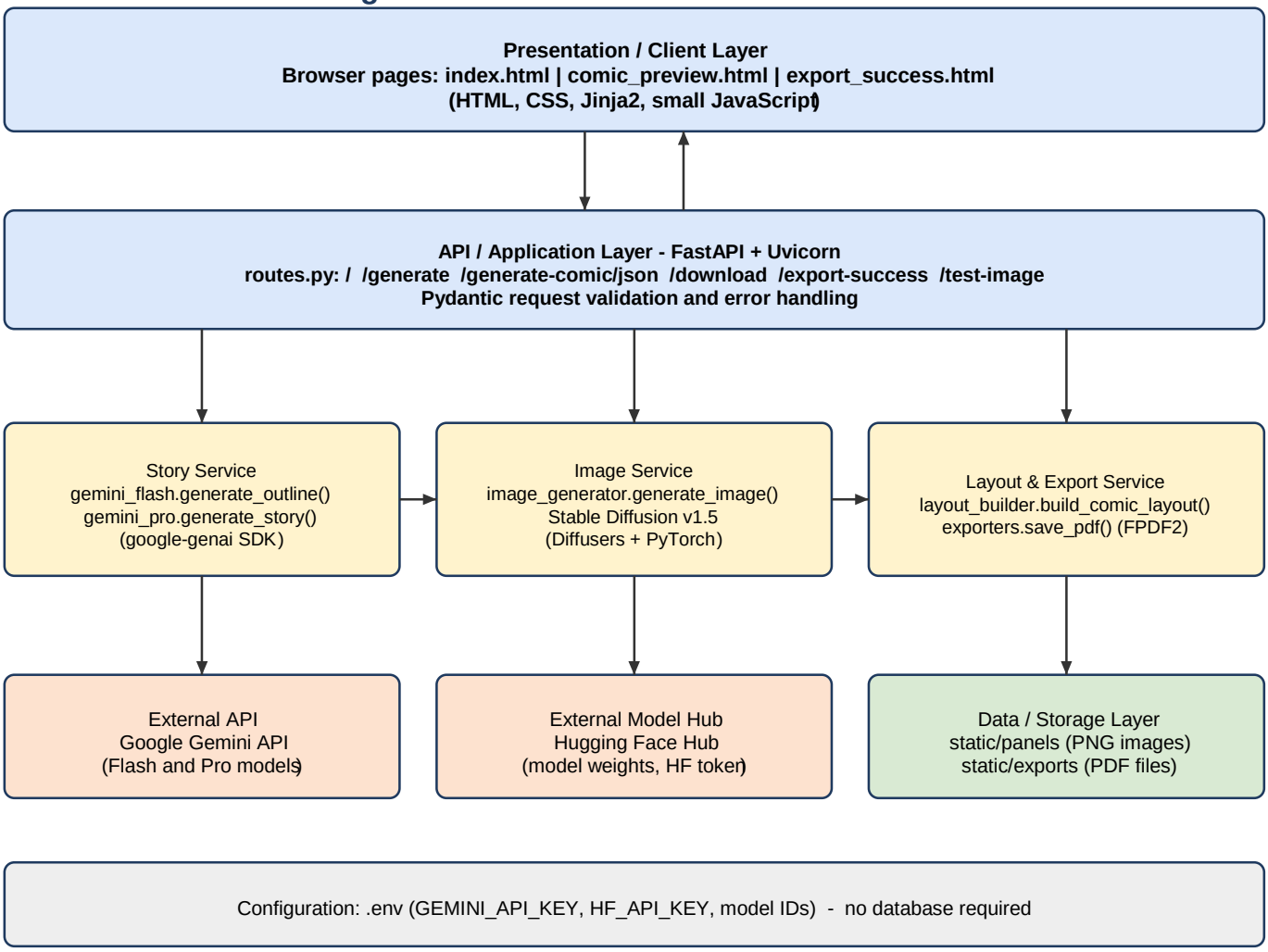
Solution Architecture

Date	30 September 2026
Team ID	3
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects story prompt, character, setting, tone and art style; shows the panel-by-panel preview, download button and success page.	HTML, CSS, Jinja2 templates, small JavaScript
API / Application Layer	Routes requests, validates input (Form / PromptRequest), orchestrates the pipeline and returns HTML or JSON; handles errors.	FastAPI, Uvicorn, Pydantic

Component Name	Description / Role in Architecture	Technologies Used

Story Service	Gemini Flash creates the structured 5-panel outline (JSON); Gemini Pro writes narration and dialogue; retries and JSON cleaning included.	google-genai SDK, Gemini Flash and Pro
Image Service	Generates one comic-style image per panel and saves it; falls back to a placeholder panel if the model is unavailable.	Hugging Face Diffusers, PyTorch, Stable Diffusion v1.5, Pillow
Layout & Export Service	Matches images with panel text into a layout and compiles the multi-page timestamped PDF.	Python, FPDF2
Storage Layer	Holds generated images and PDFs on disk; secrets stay in .env. No database is needed.	Local file system, python-dotenv